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3.	Write a Python program to calculate the area and perimeter (circumference) of a circle given its radius.		
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12.	Write a python program to create a list of five students by taking input from user and to display names of the students in ascending and descending order.		
13.	Write a Python program to take a string as input and convert the string into uppercase, lowercase, and title format.		
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16.	Write a Python program to create a tuple of five integers and apply len(), max(), min(), and sum() functions on tuple.		
17.	Write a Python program that defines two functions: • **area(l, b)** to calculate the area of a rectangle • **peri(l, b)** to calculate the perimeter of a rectangle and accept the length and breadth of the rectangle from the user and compute the area and perimeter using these functions.		
18.	Write a Python program to create a function **calc(a, b)** that takes two numbers as input and returns a list containing the results of addition, subtraction, multiplication, and division of the two numbers.		
19.	Write a Python program that imports a module mycalc and uses its functions to perform basic arithmetic operations: addition, subtraction, multiplication, and division. The program should take two numbers as input and print the results of each operation.		
20.	Write a Python program to perform division operation on two integers and also handle ValueError and ZeroDivisionError.		
21.	Write a Python program to open a file for read operation and also handle FileNotFoundError.		
22.	Write a Python program to create a user-defined package containing two modules: one for performing addition and another for performing multiplication. Import the package in the main program and use the functions to add and multiply two numbers entered by the user.		
23.	Write a Python program to create a NumPy array containing the marks of five students and display them as a **bar chart** using the Matplotlib library.		
24.	Create a window having two radio button, one label with male and second with Female and create a button and call a function name show_msg() which displays a message using messageBox.		
25.	Write a Python program to create a class with the name **"MyClass"**. In MyClass create a function **"sayHello()"** which receives the name of the user as a parameter and prints **"Hello, name!"**.		

Program-1

Program Objective: Write a python program to print a message (Hello, Welcome!) with your name and address on screen.

Program :

```
name = input("Enter your name: ")
address = input("Enter your address: ")
print("Hello, Welcome!")
print(name)
print(address)
```

Output :

```
Enter your name: Avni Sharma
Enter your address: Hapur
Hello, Welcome!
Avni Sharma
Hapur
```

Program-2

Program Objective: Write a Python program that takes two numbers as input from the user and calculates the following operations: addition, subtraction, multiplication, and division.

Program :

```
num1 = float(input("Enter the first number: "))
num2 = float(input("Enter the second number: "))
add = num1 + num2
sub = num1 - num2
mul = num1 * num2
div = num1 / num2 if num2 != 0 else "undefined (division by zero)"
print("Addition:", add)
print("Subtraction:", sub)
print("Multiplication:", mul)
print("Division:", div)
```

Output :

```
Enter the first number: 24
Enter the second number: 16
Addition: 40.0
Subtraction: 8.0
Multiplication: 384.0
Division: 1.5
```

Program-3

Program Objective: Write a Python program to calculate the area and perimeter (circumference) of a circle given its radius.

Program :

```
rad = float(input("Enter the radius of the circle: "))
Area = (22/7)*rad**2
Perimeter = 2*(22/7)*rad
print("Area of the circle:", Area)
print("Perimeter of the circle:", Perimeter)
```

Output :

```
Enter the radius of the circle: 7
Area of the circle: 154.0
Perimeter of the circle: 44.0
```

Program-4

Program Objective: Write a Python program to calculate the volume and surface area of a cuboid based on the given dimensions: length, breadth, and height.

Program :

```
l = float(input("Enter the length of the cuboid: "))
b = float(input("Enter the breadth of the cuboid: "))
h = float(input("Enter the height of the cuboid: "))
v = l*b*h
sa = 2*(l*b+b*h+h*l)
print("Volume of the cuboid:", v)
print("Surface area of the cuboid:", sa)
```

Output :

```
Enter the length of the cuboid: 6
Enter the breadth of the cuboid: 4
Enter the height of the cuboid: 5
Volume of the cuboid: 120.0
Surface area of the cuboid: 148.0
```

Program-5

Program Objective: Write a python program to check given number is even or odd.

Program :

```
num = int(input("Enter a number: "))
if num%2==0:
   print(num,"is an even number.")
else:
   print(num,"is an odd number.")
```

Output :

```
Enter a number: 56
56 is an even number.
```

Program-6

Program Objective: Write a Python program to find a greatest number in given three integers.

Program:

```
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))
c = int(input("Enter third number: "))
if a>b and a>c:
   print("The greatest number is:", a)
elif b>a and b>c:
   print("The greatest number is:", b)
else:
   print("The greatest number is:", c)
```

Output :

```
Enter first number: 34
Enter second number: 65
Enter third number: 23
The greatest number is: 65
```

Program-7

Program Objective: Write a Python program to print the multiplication table of a given number.

Program:

```
num = int(input("Enter a number: "))
for i in range(1, 11):
   print(num, "x", i, "=", num*i)
```

Output:

```
Enter a number: 5
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50
```

Program-8

Program Objective: Write a python program to print factorial of given number.

Program:

```
Num = int(input("Enter a number: "))
fact = 1
for i in range(1,num+1):
   fact = fact*i
print("The factorial of",num,"is:",fact)
```

Output:

```
Enter a number: 5
The factorial of 5 is: 120
```

Program-9

Program Objective: Write a python program to check given number is prime or not.

Program:

```
num = int(input("Enter a number: "))
if (num == 1):
   print(num,"is not a prime number.")
else:
   for i in range(2,num):
      if(num % i == 0):
         print(num,"is not a prime number.")
         break
   else:
      print(num,"is a prime number.")
```

Output:

```
Enter a number: 7
7 is a prime number.
```

Program-10

Program Objective: Write a Python program to create a list of ten integers by taking input from the user and print the Sum and Average of numbers.

Program:

```
n = int(input("Enter the number of integers you want to input: "))
lst1 = []
for i in range(n):
    num = int(input("Enter a number: "))
    lst1.append(num)
print("Sum of the numbers is:", sum(lst1))
print("Average of the numbers is:", sum(lst1)/n)
```

Output:

```
Enter the number of integers you want to input: 5
Enter a number: 23
Enter a number: 35
Enter a number: 42
Enter a number: 58
Enter a number: 64
Sum of the numbers is: 222
Average of the numbers is: 44.4
```

Program-11

Program Objective: Write a Python program to create a list of ten integers by taking input from the user and check whether a given number is present in the list or not.

Program:

```
n = int(input("Enter the number of integers you want to input: "))
lst1 = []
for i in range(n):
    num = int(input("Enter a number: "))
    lst1.append(num)
num = int(input("Enter a number to check: "))
if num in lst1:
    print(num,"is present in the list.")
else:
    print(num,"is not present in the list.")
```

Output:

```
Enter the number of integers you want to input: 5
Enter a number: 82
Enter a number: 56
Enter a number: 73
Enter a number: 48
Enter a number: 52
Enter a number to check: 73
73 is present in the list.
```

Program-12

Program Objective: Write a python program to create a list of five students by taking input from user and to display names of the students in ascending and descending order.

Program:

```
n = int(input("Enter the number of students: "))
students = []
for i in range(n):
    name = input("Enter the name of student {}: ".format(i + 1))
    students.append(name)
print("Students in ascending order:")
for student in sorted(students):
    print(student)
print("Students in descending order:")
for student in sorted(students, reverse=True):
    print(student)
```

Output:

```
Enter the number of students: 4
Enter the name of student 1: Avni
Enter the name of student 2: Bhakti
Enter the name of student 3: Astha
Enter the name of student 4: Avika
Students in ascending order:
Astha
Avika
Avni
Bhakti
Students in descending order:
Bhakti
Avni
Avika
Astha
```

Program-13

Program Objective: Write a Python program to take a string as input and convert the string into uppercase, lowercase, and title format.

Program:

```
str1 = input("Enter a string: ")
upper_str = str1.upper()
lower_str = str1.lower()
title_str = str1.title()
print("Uppercase:", upper_str)
print("Lowercase:", lower_str)
print("Title case:", title_str)
```

Output:

```
Enter a string: Hello, good morning!
Uppercase: HELLO, GOOD MORNING!
Lowercase: hello, good morning!
Title case: Hello, Good Morning!
```

Program-14

Program Objective: Write a Python program to take a sentence as input and count the number of words in the given sentence.

Program:

```
sentence = input("Enter a sentence: ")
words = sentence.split()
word_count = len(words)
print("The number of words in the given sentence is:", word_count)
```

Output:

```
Enter a sentence: Hello, How are you?
The number of words in the given sentence is: 4
```

Program-15

Program Objective: Write a Python program to take a string as input and check given string is Palindrome or not.

Program:

```
str1 = input("Enter a string: ")
str2 = str1[ : :-1]
if str1 == str2:
   print("The given string is a Palindrome.")
else:
   print("The given string is not a Palindrome.")
```

Output:

```
Enter a string: deified
The given string is a Palindrome.
```

Program-16

Program Objective: Write a Python program to create a tuple of five integers and apply len(), max(), min(), and sum() functions on tuple.

Program:

```
t = ()
for i in range(5):
    num = int(input("Enter an integer: "))
    t += (num,)
print("Tuple:", t)
print("Length of tuple:", len(t))
print("Maximum value in tuple:", max(t))
print("Minimum value in tuple:", min(t))
print("Sum of all elements in tuple:", sum(t))
```

Output:

```
Enter an integer: 12
Enter an integer: 28
Enter an integer: 34
Enter an integer: 46
Enter an integer: 50
Tuple: (12, 28, 34, 46, 50)
Length of tuple: 5
Maximum value in tuple: 50
Minimum value in tuple: 12
Sum of all elements in tuple: 170
```

Program-17

Program Objective: Write a Python program that defines two functions:

• area(l, b) to calculate the area of a rectangle

• peri(l, b) to calculate the perimeter of a rectangle

and accept the length and breadth of the rectangle from the user and compute the area and perimeter using these functions.

Program:

```
def area(l,b):
    Area=l*b
    print("Area of Rectangle is:",Area)
def peri(l,b):
    Peri=2*(l+b)
    print("Perimeter of Rectangle is:",Peri)
l=int(input("Enter Length: "))
b=int(input("Enter Breadth: "))
area(l,b)
peri(l,b)
```

Output:

```
Enter Length: 35
Enter Breadth: 25
Area of Rectangle is: 875
Perimeter of Rectangle is: 120
```

Program-18

Program Objective: Write a Python program to create a function calc(a, b) that takes two numbers as input and returns a list containing the results of addition, subtraction, multiplication, and division of the two numbers.

Program:

```
a=int(input("Enter a number: "))
b=int(input("Enter another number: "))
def calc(a,b):
    add = a+b
    sub = a-b
    mul = a*b
    div = a/b if b!=0 else "undefined (division by zero)"
    return [add, sub, mul, div]
print(calc(a,b))
```

Output:

```
Enter a number: 25
Enter another number: 4
[29, 21, 100, 6.25]
```

Program-19

Program Objective: Write a Python program that imports a module mycalc and uses its functions to perform basic arithmetic operations: addition, subtraction, multiplication, and division. The program should take two numbers as input and print the results of each operation.

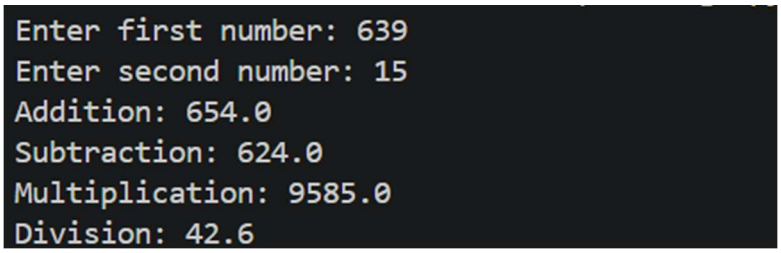
Module - mycalc.py:

```
def add(a, b):
   return a + b
def subtract(a, b):
   return a – b
def multiply(a, b):
   return a * b
def divide(a, b):
   if b == 0:
      return "Cannot divide by zero"
   return a / b
```

Program:

```
import mycalc
num1 = float(input("Enter first number: "))
num2 = float(input("Enter second number: "))
print(f"Addition: {mycalc.add(num1, num2)}")
print(f"Subtraction: {mycalc.subtract(num1, num2)}")
print(f"Multiplication: {mycalc.multiply(num1, num2)}")
print(f"Division: {mycalc.divide(num1, num2)}")
```

Output:


Enter first number: 639
Enter second number: 15
Addition: 654.0
Subtraction: 624.0
Multiplication: 9585.0
Division: 42.6

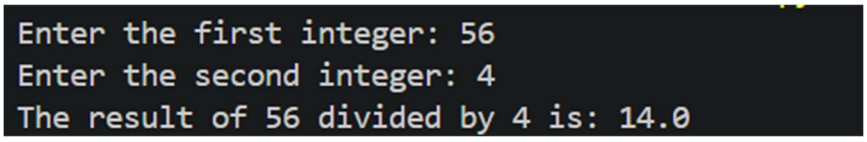
Program-20

Program Objective: Write a Python program to perform division operation on two integers and also handle ValueError and ZeroDivisionError.

Program:

```
try:
    num1 = int(input("Enter the first integer: "))
    num2 = int(input("Enter the second integer: "))
    result = num1 / num2
    print(f"The result of {num1} divided by {num2} is: {result}")
except ValueError:
    print("Error: Please enter valid integers.")
except ZeroDivisionError:
    print("Error: Second integer cannot be zero.")
```

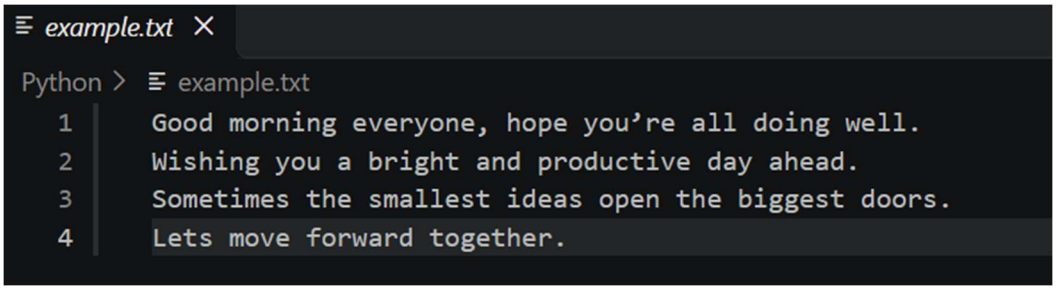
Output:



Program-21

Program Objective: Write a Python program to open a file for read operation and also handle FileNotFoundError.

Content of file - example.txt:



Program:

```
try:
    file = open("Python/example.txt", "r")
    content = file.read()
    print("File content: ")
    print(content)
except FileNotFoundError:
    print("Error: File not found.")
```

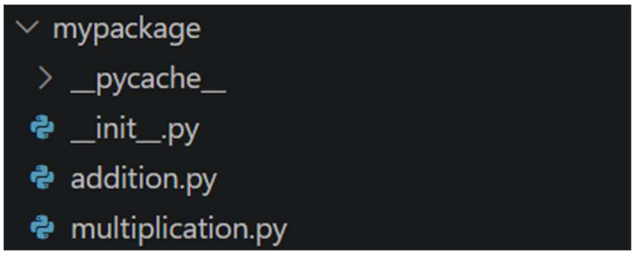
Output:

```
File content:
Good morning everyone, hope youâ€™re all doing well.
Wishing you a bright and productive day ahead.
Sometimes the smallest ideas open the biggest doors.
Lets move forward together.
```

Program-22

Program Objective: Write a Python program to create a user-defined package containing two modules: one for performing addition and another for performing multiplication. Import the package in the main program and use the functions to add and multiply two numbers entered by the user.

Folder Structure:



Program:

- **__init__.py**

```
from .addition import add
from .multiplication import multiply
```

- **addition.py**

```
def add(a, b):
    return a + b
```

- **multiplication.py**

```
def multiply(a, b):
    return a * b
```

- **main.py**

```
from mypackage import add, multiply
def main():
    num1 = float(input("Enter first number: "))
    num2 = float(input("Enter second number: "))
    sum_result = add(num1, num2)
    mul_result = multiply(num1, num2)
    print(f"Addition of {num1} and {num2}: {sum_result}")
    print(f"Multiplication of {num1} and {num2}: {mul_result}")
if __name__ == "__main__":
    main()
```

Output:

```
Enter first number: 34
Enter second number: 56
Addition of 34.0 and 56.0: 90.0
Multiplication of 34.0 and 56.0: 1904.0
```

Program-23

Program Objective: Write a Python program to create a NumPy array containing the marks of five students and display them as a bar chart using the Matplotlib library.

Program:

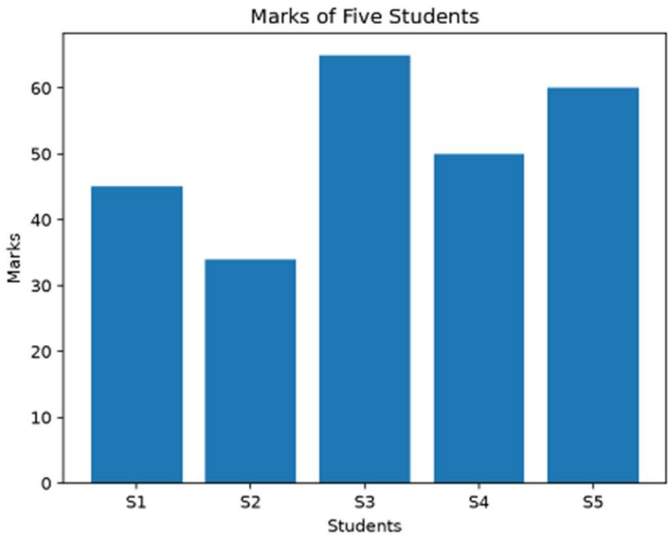
```
import numpy as np
import matplotlib.pyplot as plt
students = ['S1', 'S2', 'S3', 'S4', 'S5']
marks_list = []
for student in students:
    score = float(input(f"Enter marks for {student}: "))
    marks_list.append(score)
marks = np.array(marks_list)
plt.bar(students, marks)
plt.xlabel("Students")
plt.ylabel("Marks")
plt.title("Marks of Five Students")
plt.show()
```

Output:

Marks entered by User:

```
Enter marks for S1: 45
Enter marks for S2: 34
Enter marks for S3: 65
Enter marks for S4: 50
Enter marks for S5: 60
```

Bar Chart:



Program-24

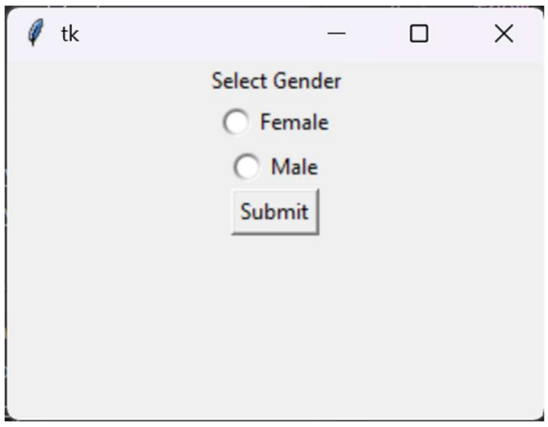
Program Objective: Create a window having two radio button, one label with male and second with Female and create a button and call a function name show_msg() which displays a message using messageBox.

Program:

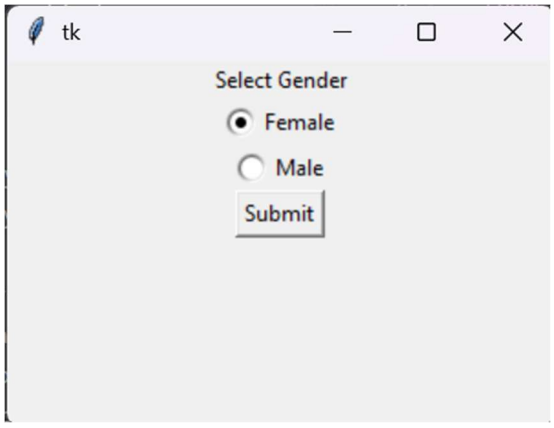
```
import tkinter as tk
from tkinter import messageBox
def show_msg():
   messagebox.showinfo("Message", "Gender selected: " + gender.get())
root = tk.Tk()
root.geometry("300x200")
gender = tk.StringVar(value=" ")
tk.Label(root, text="Select Gender").pack()
tk.Radiobutton(root, text="Female", variable=gender, value="Female").pack()
tk.Radiobutton(root, text="Male", variable=gender, value="Male").pack()
tk.Button(root, text="Submit", command=show_msg).pack()
root.mainloop()
```

Output:

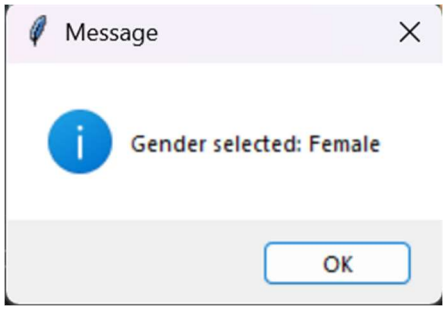
Before Selection:



After Selection:



After Clicking on Submit Button:



Program-25

Program Objective: Write a Python program to create a class with the name "MyClass". In MyClass create a function "sayHello()" which receives the name of the user as a parameter and prints "Hello, name!".

Program:

```
class MyClass:
    def sayHello(self, name):
        print(f"Hello, {name}!")
if __name__ == "__main__":
    obj = MyClass()
    name = input("Enter your name: ")
    obj.sayHello(name)
```

Output:

```
Enter your name: Avni Sharma
Hello, Avni Sharma!
```